

Team Roles of Artificial Intelligence in Anesthesiology – A Scoping Review

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Abstract

When referring to the role of newly proposed clinical applications of artificial intelligence (AI), recent work inflationary uses the term *Human-AI Team*. However, the roles foreseen for AI systems within teams remain unclear. We systematically reviewed the literature on AI deployment in anesthesiology and found that most AI system papers only describe algorithms. We identified 57 interactive systems and assigned six roles based on described behavior, tasks, and interactions. While the most prevalent role was task completer, some AI systems also served their team as problem solvers, evaluators, task motivators, or even teamwork support and team leaders. We contribute (1) a classification system for team roles, behaviors, tasks, and interactions of AI team members and (2) an overview of AI systems' team roles in anesthesiology. We conclude that (3) AI systems' intended social roles within teams need to be more consciously reflected, shaped and clearly communicated to meet healthcare standards.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**.

Keywords

Healthcare, Anesthesiology, Artificial Intelligence, Automation, Human AI Teaming

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1 Introduction

Following the advancements in artificial intelligence (AI), recent publications called for Human-AI teams in safety-critical domains [61], and particularly in acute care medicine such as anesthesiology, intensive care, and emergency care [e.g., 83]. In anesthesiology, tasks performed by the team of anesthesiologist and anesthetic nurse involve the monitoring of vital sign parameters, instrumentation, and drug administration, as well as documentation of all interventions. These task demands, which require vigilance, arithmetic computation, and memory can perfectly be supported by AI systems. Consequently, AI is not new to clinical settings and intelligent systems have already been introduced decades ago to, for instance, monitor the depth of anesthesia [62] and even to automate drug administration [43]. However, many of the proposed AI systems do not have an actual interface, and most reviews on systems deploying AI neglect the analysis of socio-technical aspects but focus on effectiveness of the system [e.g., 74, 87]. In this scoping review, we systematically analyzed the team roles assigned to AI systems based on programmed behavior, communication, and tasks that the AI systems have to do as part of the socio-technical system. We contribute 1) a classification system for AI roles in clinical teams, and 2) an overview of AI systems' team roles in anesthesiology teams. Based on the understanding of current roles, 3) we outline opportunities for AI systems in the near future and methodological paths required to design more sociable AI systems.

The literature indicates that technology not only has the potential to become a team member but is already treated as one [60]. Medical devices are no exception to being assigned social roles. For

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example, anesthesiologists' perception of their equipment includes roles such as a second patient requiring care [27] or a fellow team member that is treated with respect, given a nickname, or is being verbally calmed when it emits alarms [26, 27]. Outside of anesthesiology, seasoned physicians compared the role of a decision support tool to that of a mid-level physician, who contributes their perspective to the discussion but will not be making the decision [89].

Clearly defined and communicated team roles are an important cornerstone of team effectiveness in healthcare [12, 92]. For the definition of a human team role, Driskell et al. [11] suggested three dimensions. First, *task orientation* covers effectivity parameters such as how achievement-oriented, organized, reliable, conscientious and serious a team member approaches their task. Second, *dominance* describes a team members qualities to be active, take charge, control, direct, assert, or lead. Third, *sociability* covers a team member's warm behaviors of friendliness, support, cordiality and expressions of interest. Using these three dimensions, Driskell et al. condensed 154 role descriptions from human team role literature into 13 role clusters comprising team leader, task motivator, power seeker, critic, attention seeker, negative, social, coordinator, follower, teamwork support, evaluator, problem solver, and task completer.

Considering the roles of clinical AI systems, a qualitative review [91] of clinical AI systems already identified three primary roles that AI takes in medical settings. AI systems either 1) aid clinicians as decision support, 2) prioritize patients for the clinician, or 3) autonomously manage tasks without clinicians' oversight [91]. Viewing Zajac et al.'s AI system roles through the lens of Driskell et al.'s human team roles clusters, the *behavior* of all reviewed systems fall into either the evaluator (decision support), problem solver (decision support, prioritization), or task completer (prioritization, automation) role description. Unsurprisingly, these are the roles that have the highest correlation with team effectiveness in human teams [28] – AI systems are frequently associated with efficiency boosts [37, 85]. While Zajac et al. considered literature of all clinical disciplines in their review, their focus on the technical side was narrow, seeking only AI systems based on machine learning algorithms. In our review, we broadened the AI definition to include intelligent systems that are more intelligible than machine learning algorithms such as knowledge-based approaches but focus on anesthesiology. We seek to clarify whether team roles executed by AI in the discipline of anesthesia are limited to three Zajac et al. of the 13 team roles identified by Driskell et al. when including all intelligent systems.

In this scoping review, we systematically searched literature on AI systems in anesthesiology and gained an understanding of the current roles that AI executes in the domain of anesthesiology. Our guiding question informed by the AI's assigned *behavior*, *tasks*, and *interactions* was: Which roles does the AI take within the anesthesia team?

2 Method

To determine the current roles of AI in anesthesia, we conducted a scoping review according to the PRISMA guidelines [56, 79]. Search terms and filters were iteratively discussed and optimized by the first and second author. Because the topic under study is

addressed by human computer interaction (HCI) as well as medical research, we selected the most relevant databases for both HCI (<https://ieeexplore.ieee.org/>, <https://dl.acm.org/>) and medical research (<https://pubmed.ncbi.nlm.nih.gov/>). Our final search terms were a hybrid of [anesthesia] AND [AI] and are listed in the appendix in Table 2.

We included articles and case reports describing original research that reported on either of the following by using AI (regardless of the algorithmic approach, including machine learning as well as intelligent knowledge-based systems) published before 19 September 2024:

- Systems that are already deployed
- Prototypical developments, including early prototypes with simulated core functionality
- Algorithms created for the domain of anesthesia to support tasks in anesthetic induction or the operation room

Our exclusion criteria entailed papers that:

- Described other domains than anesthesia in human medicine
- Researched towards improving anesthesia without direct use cases for surgery, such as educational applications or exploring the pharmacokinetic effects of anesthetic agents in laboratory settings.
- Represented general introductions on AI or guidelines for medical staff on how to operate AI-driven systems
- Focused exclusively on supporting administrative tasks such as staffing
- Aimed at long-term post-operative surveillance of patients with no implication for anesthesia during surgery

From the initially identified 1486 records in September 2024, 286 articles passed our eligibility criteria, 56 of which describe how clinicians can interact with 57 presented systems (Figure 1).

For the scope of the review, we further broke down the AI system's "intended use" [91] to distinguish the three dimensions *behavior*, *task*, and *interaction*, which collectively characterize the roles of an AI. While behaviors and task orientation have been used to describe human team roles [11], we needed more detailed descriptions of an AI system's interactions in the context of their task to infer team roles, which would be assigned to human team members based on the sociability and dominance of their communication [11]. In our analysis, *behavior* refers to the AI's demeanor (i.e., outward behavior) within the team (Table 3, appendix). *Task* refers to the medical task that the AI supports (Table 4, appendix). *Interaction* describes how the AI communicates within the team (and sometimes with the patient; Table 5, appendix). We formed all levels of the three dimensions *behavior*, *task*, and *interaction* inductively during our search, except for the level "algorithm", which we predefined as one extreme of the interaction dimension. During our analysis, we gave the descriptions of AI systems' intended functionality and domain more weight than the buzzwords used by the authors for their AI systems. Finally, for determining the roles of AI systems within a team, we used the team role clusters identified by Driskell et al. as inspiration. From Driskell et al.'s 13 prototypical team roles, we identified six in the body of literature and adapted the roles to match AI team members (Table 1).

After a first rater (1st author) declared their literature table as complete, a second rater (3rd author) categorized all 57 descriptions

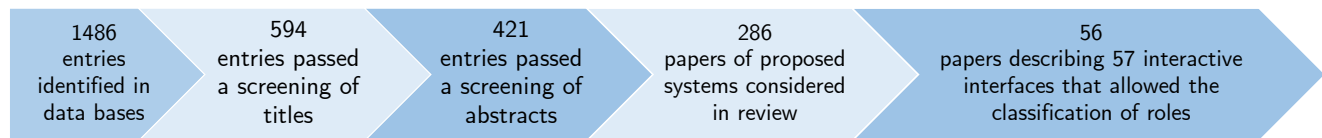


Figure 1: Selection process of literature included in our review. Note that one paper described two interfaces.

Table 1: Identified *team roles* executed by AI systems in anesthesia literature

Main role (prominent dimension)	Clarification of the role cluster adopted from Driskell et al. and adapted to suit AI systems
Evaluator (task orientation)	Inspects and evaluates and criticizes team outputs; analyses and evaluates proposals, focuses on facts and figures; team output includes current behavior (e.g., medication already administered). The evaluator includes the means of communicating their results of an evaluative process (e.g., through alerts).
Problem solver (task orientation)	Involved in the generation of ideas, problem solving; asks for information, offers facts and opinion, clarifies inconsistencies, offers solutions; provides reasoning in unexpected situations; is context aware; may act proactive but with better reasoning than the task motivator.
Task completer (task orientation)	Performs detail-oriented routine activities related to completing the task; adheres to procedures and deadlines, detail-oriented and orderly; includes routine tasks like transfer, translation, but not original solutions to rare cases.
Task motivator (dominance)	Acts as a second in command or task manager to stimulate the group to action; prods the group to action, stimulates the group, energizes the group; less reasoning than a problem solver and more motivation for the human team members to check the data themselves; includes simple task completion (i.e., checking whether a task was done or should be motivated); alarms were considered as motivating – mere system suggestions were not.
Team leader (task orientation & dominance)	Directs, organizes and leads the team toward task accomplishment; guides, brings order, convenes the group, structures the task, commands and organizes; more high-level than a problem solver, with a certain level of task motivation being expected from team lead.
Teamwork support (task orientation)	Supports team task accomplishment; pitches in to achieve team goals, implements plans, sums up and clarifies, records team activities, e.g. moderation or creating a public protocol.

of interactive systems using the dimension descriptions established by the first rater. We computed Cohen’s kappa for interrater agreement with JASP [34]. The interrater agreement for the dimension *task* reached a Cohen’s κ of .88, (95%CI, .78 to .97), and for the dimension *behavior*, Cohen’s κ was .84, (95%CI, .73 to .95), which can be considered an almost perfect agreement [40]. For the dimension *interaction*, the agreement was still substantial, with Cohen’s κ = .70, (95%CI, .57 to .84). Judging the main role of AI systems, however, yielded only fair agreement between ratings with Cohen’s κ = .25, (95%CI, .06 to .43). Some of the described systems are rather powerful and take multiple roles. When generously counting matches with secondary roles identified by either rater, the kappa score increased, but remained within the realms of fair agreement (Cohen’s κ = .37; 95%CI, .18 to .57). The most frequent inconsistency between raters was whether an AI system exceeded the role of task completion (see Table 6). In a next step, raters discussed differences and came to a consent decision regarding each dimension for each of the 57 systems. During this process, the dimension descriptions were extended to be more distinct (Table 1).

3 Results

In this subsection, we present the identified **roles** along with their typical *behaviors*, *tasks*, and *interactions* within the team. The level

of task orientation was best described in publications and, hence, had the greatest impact on systems’ role classification. One paper described two separate systems; another nine of the described systems fulfilled more than one role. Consequently, those systems occur repeatedly in the following analysis. Sorted by the frequency of AI systems’ primary role, we identified 34 Task Completers, nine Problem Solvers, five Task Motivators, five Evaluators, three Team Leaders, and one Teamwork Support. The distribution of identified primary roles from AI systems’ reported *behaviors*, *tasks*, and *interactions* can be tracked in Figure 2. The full literature table is presented in the appendix, Table 7. In the following we provide qualitative descriptions of the roles resulting from *behavior*, *tasks*, and *interactions*, and clarify the role definitions based on edge cases.

Driskell et al. label team members who perform “detail oriented, routine activities related to completing the task” (p.495) as **task completers**. An exemplary *behavior* of intelligent systems is the detection [3, 18, 20, 51, 72] of events in streams of data with optional decision support [20, 48]. The system’s *task* is to continuously monitor signals and communicate to human team members when it detects individual or multiple diagnoses – hence the strong linkage to diagnosis systems. Closed loop drug dosing systems go one step further and lend intelligent drug control [36, 68, 93] systems the authority to independently act upon their computations without

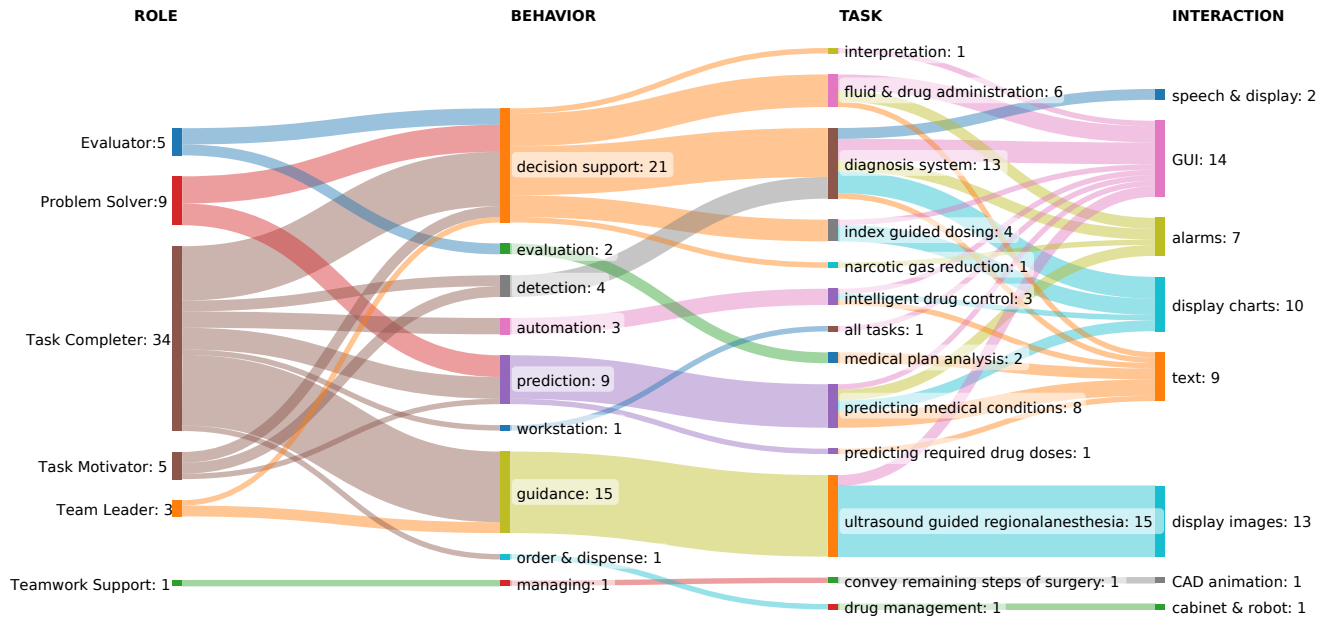


Figure 2: Overview of the main roles, behaviors, tasks and interactions of the AI systems for anesthesia in our review.

the clinician as an approving instance. Within their predefined, isolated sub tasks, those systems are given full agency and control – attributes indicative of a high level of *task* automation [66]. Another example of *task* automation to increase efficiency is the holistic order and dispense process with delivery robots [45]. The fluid administration system described by Joosten et al. has the typical high task orientation and orderliness of a task completer and offers minimalist decision support, only one button click away from being an automated closed loop system [9, 36].

Within regional anesthesia (i.e., pain medication without cardiovascular effects typically blocking nerves for entire limbs), AI enhanced systems serve as guidance, leading the clinician to the spot where the needle must be placed. The handheld ultrasound transducer serves as main input device, the AI enhances and labels the incoming image data and displays images enriched to outline anatomical features as output. However, there are two distinct roles those technical guides take, best described metaphorically. On the one side of the guidance spectrum, the majority of ultrasound guidance systems act as a magic lantern held by the clinicians to render visible what is hidden to the naked eye [4–7, 10, 23, 24, 42, 44, 47, 67, 77, 90] – a detail oriented routine activity executed on command by **task completers**. On the other side, the software described in two publications introduces a restrictive GUI, where the next anatomical landmark can only be explored after the former step was completed and met success criteria predefined in the software [32, 63]. The latter AI systems exceed vision enhancement as a task completion service to additionally structure the *task* and command clinicians on what to do. Here, the software does not only provide the magic lantern but also acts as the guide who is holding the light and, thus, the authority to proceed with the next step. Such dominant guidance *behavior* enforcing a *task* structure clearly outs those AI systems as **team leaders**.

Prediction of events is technically close to detection yet fulfills another role within the team. In contrast to an alarm sound when a threshold is passed, communicating a warning of an imminent condition 2-5 minutes ahead [49, 86] gives teams the time to observe the development of vital parameters and discuss actions. When possible postoperative states and developments [8, 16] or drug requirements [82] are predicted based on patient records before the operation begins, the role shifts further from an in-situ alarmist to an experienced advisor. Here, great differences of how systems communicate their prediction determined their team role. A mere risk calculator [8, 15] that produces the estimated failure rate as numeric text output upon request suits the definition of **task completer**. Systems that provided appropriate reasoning for their predictions [13, 16, 49, 80] suited our definition of the role **problem solver**. However, five systems in our review, among them four task completers, point out estimated trends without giving advice on how to act upon their predictions [76, 82, 86] or detections [3, 72]. Generally “prod[ding] the group to action” [11, p.495] is *behavior* indicative of a **task motivator**.

Evaluation is *behavior* merely displayed by the medical plan analysis AI ATTENDING developed by Miller [54, 55] which reviews text input about an intended anesthetic procedure and returns alternative approaches as text output. While Miller highlights the critiquing abilities of the system, he also requires it to communicate in a way that is “readable, easily understood, and diplomatic” [54, p. 364]. According to Driskell et al., the diplomatic advances representing higher sociability identify ATTENDING rather as an **evaluator** than a sole critic. In addition to smaller features that can be labeled as task completion, the decision support tools by Nanji et al. serve as last-minute **evaluator** on team *behavior* reportedly in execution. Their system emits medication warnings if the scanned label of a syringe with a drug about to be injected or an entered

dose does not match the expectation of the system. While the initial warning is a one-sentence hint, clinicians interested in the rationale behind the alert are offered a link to a more detailed argumentation. Further **evaluator** systems of actions that have already started are specialized in anticoagulant infusions [25] and the reduction of excessive fresh gas flow [58].

The aim of decision support systems is to empower clinicians to make better informed decisions than before. Most systems offer facts and solutions and, consequently, can be classified as **problem solvers** [11]. Critical abilities to contribute solutions in-situ include a) appropriate means of communication for the context of anesthesia and b) comprehensible reasoning. For instance, Ross et al. created a system where the AI continuously suggests medication and dosages to be administered and the clinician needs to acknowledge or challenge those suggestions [70]. While the system communicates via easy-to-read text instructions, it does not make the steps by its reasoning engine transparent. Therefore, Ross et al.'s system is a reliable **task completer** but hardly qualifies as a good problem solver. Segall et al. followed an ecological interface design approach to create a decision support tool diagnosis system for anesthetic crises [73]. Its focus on the reasoning ability fulfills one precondition for a good problem solver. However, in a preliminary evaluation study, the text-heavy written diagnosis and treatment advice in the GUI was updated so fast and frequently that it was difficult to read. Following our problem solver definition, Segall et al.'s AI-system would contribute even better to problem solving if it appropriately communicated its knowledge. In contrast, the AlertWatch interface [78] communicates the changing vital parameters through bright color codes and emits alerts only rarely in form of a few keywords. Both diagnosis systems [73, 78] qualify as **problem solvers** yet require the constant visual attention of a vigilant team. Furthermore, the detailed structure of the AlertWatch interface [78] as well as its tone in commanding medical personnel what to do made us classify it additionally as a **team leader**.

Binding less visual attention than the AlertWatch [78], the Smart Anesthesia Manager [38, 57] only offers advice in the form of pop-up messages whenever a variable exceeds predefined limits and has context-sensitive modules that only start up during specific orders or surgeries. In fact, the Smart Anesthesia Manager's *behavior* in one study was so discreet that in a post-usage query only one third of anesthesia team members knew about its existence [38]. We categorize the role of the Smart Anesthesia Manager as a **task motivator**, even though a reluctant one. The ordering system for blood gas analysis [52] only offers advice to reduce the frequency when requested as a side note – its main purpose is the interpretation of blood gas analysis why we classified it as **task completer**. The border to task completion is clearly crossed with index guided dosing advisors: here the systems continuously display an index number which is computed from various patient measurements [84]. Strictly following the rules associated with the index significantly improves post-operative pain for the patient [17, 53, 81]. However, modern models computing the index have been pretrained and lack transparency, thus lacking the communication ability required to serve as problem solvers in the rare situations where the index guided dosing does not work.

The system CASSANDRA, designed by Klüber et al., allowed participants to see at one glance the most likely diagnosis and other

alternative diagnoses. Designed to be a devil's advocate and critic, CASSANDRA's socially neutral form of conveying facts renders it a **problem solver** in the first place. Interestingly, users did not only rely on CASSANDRA's decision support *behavior* but additionally saw it as a pool of shared knowledge among all team members in the room. These activities of **teamwork support** [11] extend CASSANDRA's roles beyond a mere problem solver. Lee et al. prevented their multi-modal alarm system from such knowledge sharing with the entire team (and thus taking on the additional role) through restricting the audio channel to the anesthetist wearing a Bluetooth headset.

Offering evidence and treatment advice, Lee et al.'s system is an early example of a **problem solver** within the anesthesia team. Similarly, Greenberg et al. present a simulation, and Jia and Yang present the technical architecture for speech assistant systems, which do not directly interact with the entire team but rather act as a personal assistant and **problem solver** for a single clinician. A second example for **teamwork support**, however, with an entirely different *task* and *behavior* shadows the ongoing surgery process and translates its latest insights into a status [88]. The resulting animated surgery status supports the anesthetist's time management to gradually reduce anesthetics towards the end of the procedure.

Finally, a system not limited to one specific role, *task* or *behavior* is the holistic workstation, which comprises drug control, documentation, diagnosis, and smart alarms, and represents more of a virtual caregiver-sidekick than a task specific system [64]. Since the publication does not offer information on coordinative behavior or guidance, the associated roles are restricted to **task completer** and **problem solver**.

A majority of 230 papers within the 286 considered in this review (see Figure 1) described algorithmic approaches. Their *behavior* covered, for instance, constant monitoring of vital parameters [94] or making predictions about medical conditions [33], intervention duration [29], or intervention difficulty [46]. These papers generally reported the effectiveness of the algorithm but, unfortunately, it was not possible to effectively infer the team roles from reports that lack the description of an interface, because a crucial part to teamwork is how the team members communicate with each other.

4 Discussion

For this scoping review, we systematically analyzed AI systems' roles in anesthesia and found that AI systems contribute to a wide range of *tasks*, however, most work exclusively in the background without an interactive interface to the clinician. In this majority of publications, a solely algorithmic description impeded the identification of whether the AI application can be considered as a team member. Among the interactive systems, task orientation was the most prevalent dimension comprising roles such as **task completers**, **problem solvers**, **evaluators** (confirming the findings by Zajac et al.), and additionally **teamwork supporters**. Assigning computer systems team roles associated with detail-oriented *tasks* is most intuitive because tirelessly processing large amounts of data while adhering to predefined procedures in an orderly fashion is a *behavior* in which computers have always been superior to humans [14].

However, we also identified the roles **task motivator** and **team leader** which are characterized by an increased dominance and thus exceed prior findings [91]. This prompts the question whether the previous predominance of task orientation in AI systems was a result of technical limitations. While we only identified AI taking the role of a team leader in more recent literature between 2018 [78] and 2021 [32], occurrences of task motivators reach from 1991 [72] to 2024 [76]. Therefore, we assume that dominance of AI systems is not dependent on technical advances. Whether the recent emergence of AI systems adopting team leadership is a result of user needs, trust or broader societal acceptance of Human-AI teams is to be answered in future work. For systematic reviews on team roles of AI systems in other domains we encourage to maintain unrestricted time-frames in search prompts to also include the insightful early works of Human-AI teaming.

A team member dimension that is still hardly addressed by AI systems today is sociability. While it is apparent why no one would intentionally design a negative system that “judges others [...] seeks power [...] seeks attention and resources but does little work [or ...] displays negative affective behavior” [11, p. 495], the positive end of the sociability scale has not been used either. It holds potential for anesthesia teams, offering roles such as a **social** character who increases harmony and mediates in conflicts, a **coordinator** who facilitates participation within and beyond the team, and a **follower** who actively listens to ideas and participates in discussions [11]. An artificial social team member may, for instance, alleviate negative effects of hierarchies within clinical teams [21] or contribute to a positive environment for reporting of incidents and a consequential rise in patient safety [65].

Stewart et al. showed that the consistency of human team members' social roles is linked to higher task performance [75]. Therefore, the roles an AI takes within a team need to be more consciously shaped. AI systems' social abilities and how they articulate dominance can be considered in the design process [e.g., 31]. Resulting AI systems' psychometric qualities can be measured [69] and it can be evaluated whether they are perceived as teammates [e.g., 2, 50].

Only a fifth of the publications included in our review described an interface. Among those, the focus was on technical validity of task execution and less on the user experience of the deployed system (but see [39]). Put in team terminology, the AI systems' ability complemented the team, but sometimes communicational skills were neglected, thus decreasing the benefit for the team. While communication among humans is typically multi-modal and context adopted regarding its detail, the systems' communication is predominantly visual and ranges from constantly displaying an index [17, 53, 81] to only emitting event-triggered alerts [38, 57, 58]. Noteworthy exemptions that potentially hit just the right tone are the diplomatic enunciation of the medical plan analysis system ATTENDING [54] and the diagnosis system CASSANDRA which served as shared knowledge pool supporting teamwork [39]. Being included in all communication channels may raise an AI system's perception as a team member. In an environment where explicit, verbal communication is preferred [12] and effective [22], intelligent systems should master speech interaction to strengthen their team role. Whereas Lee et al. claimed in 2008 that voice recognition had not been accurate enough, this was already feasible with limitations at the time [1] and stands without discussion today [30, 35]. Still,

most decision support systems only offer advice as written text or graphical representation, binding the team's visual attention. Natural language communication which may further include an intelligent system into the team [95] was only suggested in three papers as input [39], output [41] or bi-directional interaction [35]. Whether AI systems were built to communicate with the entire team [39] versus only speaking to the anesthetist [41] was not the focus of this work. Future systematic reviews on roles in Human-AI Teams could elaborate in greater detail which human team members interact in which capacity with the AI systems.

The initially only fair agreement between raters regarding the AI's team roles indicates that human team roles could not be inferred from descriptions of AI systems as easily as their *behavior*, *task*, and *interaction*. However, clarity about team roles – best communicated explicitly – is crucial to team effectiveness in healthcare [12, 92]. With some of the AI systems in our review apparently adopting more than one role or implicitly switching roles depending on the situation the requirement for team role clarity is violated. Therefore, we call for future AI systems in healthcare to abide to the same principles as human team members for whom “team roles need to be clearly understood and enacted.” [12, p. 99]

4.1 Conclusion

In this scoping review, we identified 286 papers of AI systems in anesthesia and 57 descriptions of interactive systems that allowed for role classifications. We found the 13 team role clusters suggested by Driskell et al. useful for recognizing AIs' roles in human teams, and found four roles with high task orientation (**task completer**, **problem solver**, **evaluator**, and **task motivator**) confirming former reviews of clinician-facing AI systems [91]. Additionally, we identified **team leader** and **teamwork support** as roles with high dominance, extending the known team roles assigned to clinical AIs. Implications for future work are to award AI team members stronger capabilities on the social dimension (e.g., **social**, **follower** or **coordinator**) in order to shape more positive and thus safer working environments in healthcare. More importantly, with capable AI systems in clinical Human-AI Teams adopting multiple roles [e.g., 39, 64, 78] and interacting with all other work system entities [71] those actors should be designed to clearly communicate in each situation which roles they currently execute.

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Appendix

Table 2: Search terms for each database accessed on 19 September 2024. Note that we omitted the term “agent” in the PubMed database where it was used predominantly as “anesthetic agent” and produced over 9000 additional results.

Database	Search term
ACM	Abstract=(anaesthesia OR anesthesia OR anaesthesiology OR anesthesiology OR anesthetic OR anaesthetic) AND ("artificial intelligence" OR intelligent OR AI OR agent OR scribe OR "neural network" OR "team member" OR "machine learning" OR "decision support")
IEEE	("Abstract": "anaesthesia" OR "Abstract": "anesthesia" OR "Abstract": "anaesthesiology" OR "Abstract": "anesthesiology" OR "Abstract": "anesthetic" OR "Abstract": "anaesthetic") AND ("Abstract": "artificial intelligence" OR "Abstract": "intelligent" OR "Abstract": "AI" OR "Abstract": "agent" OR "Abstract": "neural network" OR "Abstract": "machine learning" OR "Abstract": "decision support")
PubMed	(anaesthesia[Title/Abstract] OR anesthesia[Title/Abstract] OR anesthesiology[Title/Abstract] OR anaesthesiology[Title/Abstract] OR anesthetic[Title/Abstract] OR anaesthetic[Title/Abstract]) AND ((machine learning[Title/Abstract]) OR (neural network[Title/Abstract]) OR (AI[Title/Abstract]) OR (intelligent[Title/Abstract]) OR (decision support[Title/Abstract]) OR (artificial intelligence[Title/Abstract]))

Table 3: Levels of the dimension main *behavior* describe how the AI system approaches its task

Main behavior	Clarification of the behavior levels agreed upon by both reviewers (1st and 3rd author)
Automation	Automating a task entirely, so that it runs in a closed loop without the clinician’s interference
Decision support	Reminders or reasoning that supports the clinician’s decision process
Detection	Detecting single or multiple diagnosis from processed signals
Evaluation	Evaluating an action or suggestion made by the clinician
Guidance	Guiding clinicians to the spot where they need to administer medication [spatial]
Managing	Giving an overview and orientation on upcoming processes [temporal]
Order & dispense	Managing the logistics of clinicians’ needs [spatial and temporal]
Prediction	Predicting conditions or needs before they occur, thus allowing the clinician to plan
Workstation	an entire workstation entailing a set of behaviors

Table 4: Identified levels of the *tasks* fulfilled by AI systems in anesthesia literature

Task	Clarification of the task levels agreed upon by both reviewers (1st and 3rd author)
All tasks	Including drug control, documentation, diagnosis, smart alarms.
Progress indication	Convey remaining steps of surgery
Diagnosis system	Managing drug availability before they enter the patient
Drug management	Managing drug availability before they enter the patient
Fluid & drug administration	Reminding the clinician to administer fluids, drugs, nutrients
Index guided dosing	Administering drugs according to an index computed by the intelligent system
Intelligent drug control	An intelligent system autonomously administering medication
Interpretation	E.g., of blood gas analysis
Medical plan analysis	Analyzing the medical plan before it is executed (e.g., interventions, procedure)
Narcotic gas reduction	Reducing the amount of gas the patient is exposed to (to reduce side effects and environmental impact)
Predicting medical conditions	E.g., prompting scores or probabilities for likely conditions
Predicting required drug doses	E.g., prompting estimations for liquids, glucose, or hypnotics to be administered
Ultrasound guided regional anesthesia	Helping clinicians to find the perfect spot for injections to reduce the amount of puncture attempts and unintended tissue damage

Table 5: Levels of the dimension *interaction* describe how the AI system communicates with the team

Main interaction	Clarification of the interaction levels agreed upon by both reviewers (1st and 3rd author)
Algorithms	Only the algorithm of an AI system but no interface is available or described
Alarms	Pop-ups and multi-modal alarms
Cabinet & robot	e.g., an automated logistic system for drug provision
CAD animation	Computer Aided Design (CAD) animation (of a patient's anatomy)
Display charts	Fuzzy charts, curves, vital parameters, and indices
Display images	With or without color overlays
GUI	Ranging from multi-page forms to Graphical User Interfaces (GUI) with additional audio
Speech & display	Multi-modal systems communicating via voice and displays
Text	e.g., forms, dynamically generated checklists, question forms, tables

Table 6: Contingency table showing agreement of first rater (1st author) and second rater (3rd author) on the roles of AI systems in human-AI teams

Second rater	First rater					Total
	evaluator	problem solver	task completer	task motivator	team leader	
evaluator	2	1	3	0	0	6
problem solver	0	5	0	1	0	6
task completer	0	3	32	1	2	38
task motivator	0	2	3	0	0	5
team leader	0	0	1	0	0	1
teamwork support	0	0	1	0	0	1
Total	2	11	40	2	2	57

Table 7: The Literature table is sorted alphabetically by role. Where multiple roles or tasks are listed the primary role or task is named first.

Reference	Role	Behavior	Task	Interaction
Gupta [25]	Evaluator	decision support	fluid & drug administration	alarms
Nair et al. [58]	Evaluator	decision support	narcotic gas reduction, drug administration	alarms
Miller [54]	Evaluator	evaluation	medical plan analysis	text
Miller [54]	Evaluator	evaluation	medical plan analysis	text
Nanji et al. [59]	Evaluator, Task Completer	decision support	fluid & drug administration	GUI
Lundberg et al. [49]	Problem Solver	prediction	predicting medical conditions	display charts
Fritz et al. [16]	Problem Solver	prediction	predicting medical conditions	GUI
Jia and Yang [35]	Problem Solver	decision support	diagnosis system	speech & display
Segall et al. [73]	Problem Solver	decision support	diagnosis system	text
Klüber et al. [39]	Problem Solver, Teamwork Support	decision support	diagnosis system	GUI
Ferrari et al. [13]	Problem Solver	prediction	predicting medical conditions	text
Greenberg et al. [19]	Problem Solver	decision support	diagnosis system	speech & display
Turan et al. [80]	Problem Solver	prediction	predicting medical conditions	text
Liu et al. [45]	Task Completer	order & dispense	drug management	cabinet & robot
Joosten et al. [36]	Task Completer	automation	intelligent drug control	display charts
Lowe et al. [48]	Task Completer	decision support	diagnosis system	display charts
van der Wal et al. [81]	Task Completer	decision support	index guided drug dosing	display charts
Fuica et al. [17]	Task Completer	decision support	index guided drug dosing	display charts
Meijer et al. [53]	Task Completer	decision support	index guided drug dosing	display charts
Mansoor Baig et al. [51]	Task Completer	detection	diagnosis system	display charts
Tang et al. [77]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Gungor et al. [24]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Compagnone et al. [10]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Yang et al. [90]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Bowness et al. [5]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Gungor et al. [23]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Bowness et al. [7]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Bowness et al. [6]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Bowness et al. [4]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Li et al. [42]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Liu et al. [44]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Paris and Hafiane [67]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Liu and Cheng [47]	Task Completer	guidance	ultrasound guided regional anesthesia	display images
Zaouter et al. [93]	Task Completer	automation	intelligent drug control	GUI
Joosten et al. [36]	Task Completer	decision support	fluid & drug administration	GUI
Wadgaonkar and Bhole [84]	Task Completer	decision support	index guided drug dosing	GUI
Meidani et al. [52]	Task Completer	decision support	interpretation	GUI
Gohil et al. [18]	Task Completer	detection	diagnosis system	GUI
Pawade et al. [68]	Task Completer	automation	intelligent drug control	text
Ross et al. [70]	Task Completer	decision support	fluid & drug administration	text
Chang et al. [8]	Task Completer	prediction	predicting medical conditions	text
Fritz et al. [15]	Task Completer	prediction	predicting medical conditions	display charts
Coeckelenbergh et al. [9]	Task Completer	decision support	fluid & drug administration	GUI
Greenhow et al. [20]	Task Completer, Problem Solver	decision support	diagnosis system	GUI
Ohligs et al. [64]	Task Completer, Problem Solver	workstation	all tasks	GUI
Wijnberge et al. [86]	Task Completer, Task Motivator	prediction	predicting medical conditions	alarms

Continued on next page

Reference	Role	Behavior	Task	Interaction
Velagapudi et al. [82]	Task Completer, Task Motivator	prediction	predicting required drug dosis	text
Kiatchai et al. [38]	Task Motivator	decision support	diagnosis system	alarms
Szrama et al. [76]	Task Motivator	prediction	predicting medical conditions	alarms
Lee et al. [41]	Problem Solver	decision support	diagnosis system	alarms
Nair et al. [57]	Task Motivator	decision support	fluid & drug administration	alarms
Becker et al. [3]	Task Motivator, Task Completer	detection	diagnosis system	display charts
Schecke et al. [72]	Task Motivator, Task Completer	detection	diagnosis system	display charts
Oh et al. [63]	Team Leader	guidance	ultrasound guided regional anesthesia	GUI
In Chan et al. [32]	Team Leader	guidance	ultrasound guided regional anesthesia	GUI
Tremper et al. [78]	Team Leader, Problem Solver	decision support	diagnosis system	GUI
Xu et al. [88]	Teamwork Support	managing	progress indication	CAD animation